



TELECOM PHYSIQUE STRASBOURG

Illumination beams for a light-sheet microscope

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Chapter 1

Introduction

Since 1960, notably with the creation of the first LASER by the American physicist Theodore Maiman on May 16 of that same year at the Hughes Research Laboratory, light technologies have continuously gained importance in many fields of engineering, making them almost ubiquitous in daily life. The study of light is particularly a key element in the field of microscopy. In this project, we focus solely on the illumination beams used in a specific type of microscope: light-sheet microscopes.

Their main field of application is the three-dimensional imaging of living specimens, as these microscopes are ideal for fast and low photo-toxic imaging: the exposure to light is short and efficient enough not to damage the sample, which may react to light. The principle of Light-Sheet Fluorescence Microscopy (LSFM) is as follows. A light beam (often Gaussian) is sent through a lens whose purpose is to flatten the output beam. The output beam thus resembles a sheet of light, which cross-sections the sample in a given plane. In this way, an entire selected slice of the sample is processed, rather than the entire sample as in epifluorescence microscopes, which justifies the precision, efficiency, and speed of this method. Subsequently, in a direction orthogonal to the plane of the light sheet, a sensor detects the light emitted by fluorescence, a consequence of spontaneous emission after photons are absorbed by the slice of the sample traversed by the light sheet.

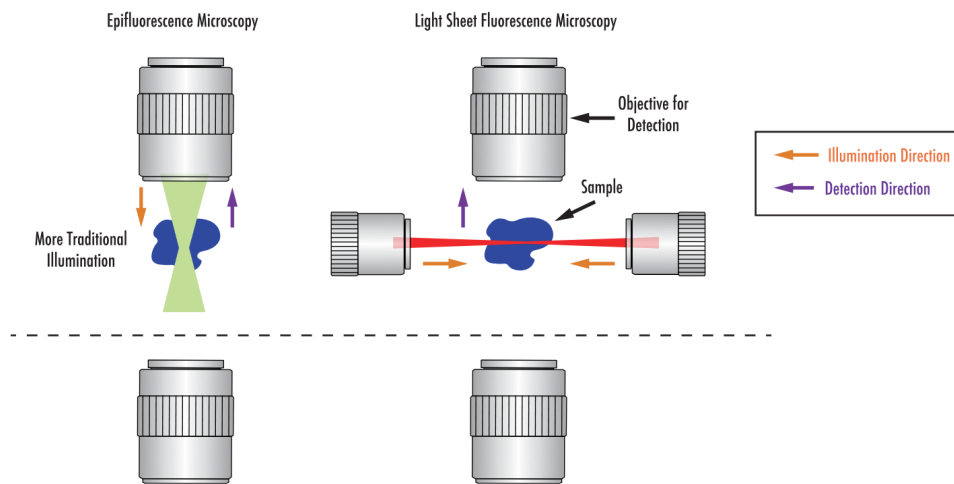


Figure 1.1: Illustration from <https://www.edmundoptics.fr/knowledge-center/application-notes/microscopy/light-sheet-fluorescence-microscopy/>

Thus, by performing numerous observations for different planes of the sample, one can reconstruct a 3D image and observe what is happening inside it. One of the key ideas of our project is that the shape of the light sheet can be changed by changing the shape of the beam that will

traverse the lens responsible for flattening it. To do this, it is necessary to view light in its wave aspect. Indeed, light, like any wave, is subject to the phenomenon of diffraction.

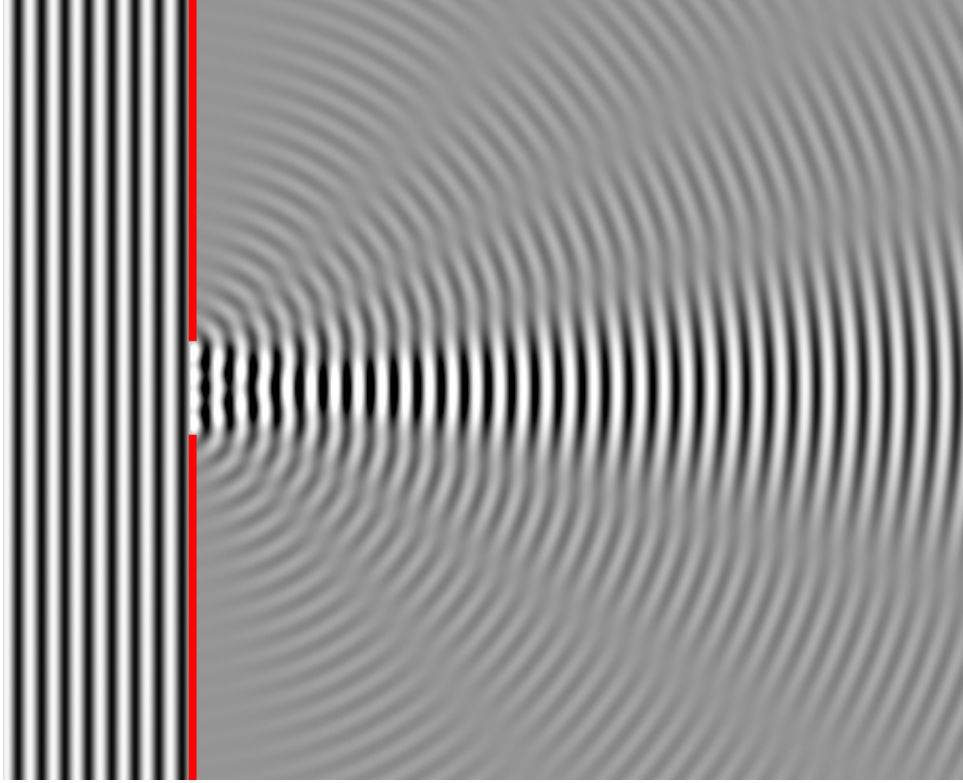


Figure 1.2: Illustration from <https://en.wikipedia.org/wiki/Diffraction>

Consequently, it is possible to modify the shape of an initially Gaussian beam by changing the shape of the pupil. Indeed, the passage of light through a hole causes the phenomenon of diffraction. This phenomenon is mathematically translated by the fact that the incident light wave can be decomposed into an infinite sum of plane waves thanks to a spatial Fourier transform. This spatial Fourier transform provides information about the amplitude and phase of each component of a plane wave defined by a given wave vector k . To modify the shape of the light sheet, we proceed as follows. We emit a Gaussian beam that passes through a device equipped with a pupil. This pupil aims to modify the shape of the beam through diffraction. We then recenter the diffracted beam by passing it through an optical system. The final beam then respects the equation:

$$\vec{E}(x, y, z) = -\frac{if}{\lambda_0 k_t^2} \mathcal{F} \left[\frac{\vec{E}_t(\theta, \phi)}{\cos(\theta)} e^{ikz} \right] \quad (1.1)$$

Finally, the resulting final beam is passed through the lens intended to flatten the beam to obtain the desired light sheet.

Project Objectives

The central objective of this project is to design, study, and compare, via numerical simulation, different illumination strategies for light-sheet microscopy (*Light Sheet Fluorescence Microscopy* - LSFM). In order to optimize three-dimensional imaging of living specimens, our work is structured around the following development axes:

- **Development of an optical propagation simulator:** Implementation under a Python environment of an algorithm based on the Fast Fourier Transform (FFT) and the Fraunhofer formalism. This simulator allows for the accurate modeling of the electric field's transition from the objective lens pupil to the focal plane.
- **Characterization of the resolution / depth of field trade-off:** Quantitative analysis of the relationship between the numerical aperture (NA) and the beam width at half maximum ($FWHM$). This study aims to define the theoretical and practical limits of the Gaussian beam, used here as a reference standard.
- **Phase engineering and robustness study:** Exploration of complex wavefronts (Bessel and Airy beams) to overcome diffraction. The study focuses on energy transport properties (*Poynting vectors*) and the self-healing capacity (*self-healing*) of beams when facing obstacles, thus simulating imaging conditions within thick or inhomogeneous biological tissues.

Chapter 2

Theory

2.1 Framework of the Study

We consider an aperture Σ contained in the $z = 0$ plane with origin O . The wave incident on the aperture will be a plane wave throughout the study and its source will be located on the z axis for $z < 0$, making its amplitude constant over the entire wavefront.

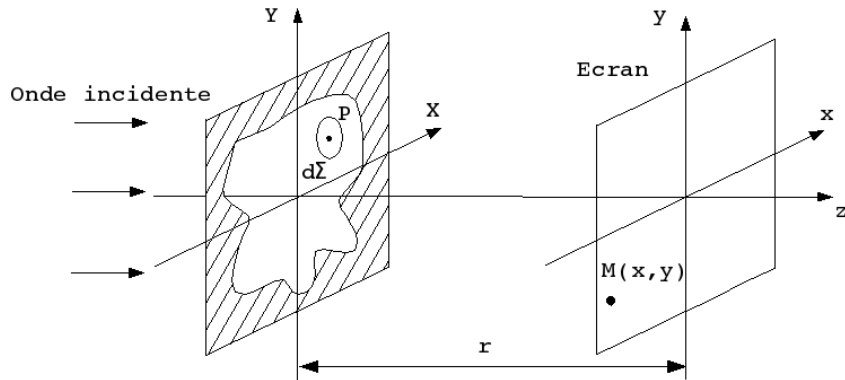


Figure 2.1: Representative diagram of the situation

Subsequently, we will be led to have other types of incident waves. To do this, we introduce $t(P) \in [0, 1]$ as the transmission factor which accounts for the transparency of the aperture defined for any point P of the $z = 0$ plane. This factor will allow modeling the same case as a wave other than a plane wave for a fully transparent aperture .

2.2 Diffraction

Let $E(P) = E(X, Y) = E_0 t(P)$ be the amplitude of the incident wave at any point P of the aperture (hence after its passage), with coordinates (X, Y) and E_0 its amplitude before the aperture. The amplitude of the wave emitted by the secondary source of surface $d\Sigma$ around P is of the form $KE(P)d\Sigma$ where K is a constant that is not necessary to determine here. When it arrives at the observation point M , with coordinates (x, y) in the plane $z = r$ with origin O' , this wave has the amplitude, in complex notation:

$$dE(M) = \frac{1}{PM} KE(P) d\Sigma e^{i\phi_{PM}} = \frac{1}{PM} KE(P) d\Sigma e^{i\frac{2\pi PM}{\lambda}}$$

the factor $\frac{1}{PM}$ accounts for the attenuation of the spherical wave emitted at P and ϕ_{PM} represents the phase shift of the wave between P and M .

The total amplitude at M is obtained by summing the contributions of all points of the aperture, namely:

$$E(M) = K \iint_{\Sigma} \frac{E(P)}{PM} e^{i\frac{2\pi PM}{\lambda}} d\Sigma$$

2.3 Fresnel Diffraction

We place ourselves in the classical observation case $d \ll r$ where d is the characteristic distance of the problem, i.e., the size of the aperture and the characteristic distance of diffraction.

Knowing that $\overrightarrow{PM} = -\overrightarrow{OP} + \overrightarrow{OO'} + \overrightarrow{O'M} = \begin{pmatrix} x - X \\ y - Y \\ r \end{pmatrix}$ according to Chasles' relation.

We obtain

$$\begin{aligned} PM &= \sqrt{r^2 + (x - X)^2 + (y - Y)^2} \\ &= r \sqrt{1 + \frac{(x - X)^2}{r^2} + \frac{(y - Y)^2}{r^2}} \\ &\text{and by Taylor expansion we obtain} \\ &\simeq r \left[1 + \frac{(x - X)^2}{2r^2} + \frac{(y - Y)^2}{2r^2} \right] \end{aligned}$$

By replacing PM in the previous expression, we obtain:

$$E(M) = \frac{KE_0 e^{i\frac{2\pi r}{\lambda}}}{r} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} t(X, Y) \exp \left[\frac{i\pi}{\lambda r} [(x - X)^2 + (y - Y)^2] \right] dX dY \quad (2.1)$$

The term $\frac{1}{PM}$ is approximated as $\frac{1}{r}$; we can afford this because it is not a periodic function, unlike the complex exponential which has a much greater sensitivity to variations in PM due to its multiplication with $\frac{1}{\lambda}$, a very large term that increases sensitivity to variations in PM .

Subsequently, we will denote $K' = KE_0 \frac{e^{i\frac{2\pi r}{\lambda}}}{r}$

Equation (2.1) is called the Fresnel transform, which allows determining the diffraction pattern at a finite distance, as long as $d \ll r$.

2.4 Fraunhofer/Far-Field Diffraction

Here we place ourselves under the conditions allowing the following approximation:

$$\exp \left[\frac{i\pi}{\lambda r} (x - X)^2 \right] \simeq \exp \left[\frac{i\pi}{\lambda r} x^2 \right] \exp \left[-\frac{i2\pi}{\lambda r} xX \right]$$

In other words, under this condition: $\frac{X^2}{\lambda r} \ll 1 \quad \forall X \in \Sigma$. The same reasoning applies to y and Y .

This finally gives us:

$$\begin{aligned} E(M) &= K' E_0 \exp \left[\frac{i\pi}{\lambda r} (x^2 + y^2) \right] \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} t(X, Y) \exp \left[\frac{i2\pi}{\lambda r} (xX + yY) \right] dX dY \\ E(M) &= \frac{KE_0}{r} \exp \left[\frac{i\pi}{\lambda r} (x^2 + y^2) \right] \hat{t} \left(\frac{xk}{r}, \frac{yk}{r} \right) e^{ikr} \end{aligned} \quad (2.2)$$

With $\hat{t}$ the Fourier transform of t and $k = \frac{2\pi}{\lambda}$ the wave vector. Physically, we find that the amplitude of the diffracted wave depends only on the direction of observation defined by the angles $\frac{x}{r}$ and $\frac{y}{r}$, which justifies the name far-field diffraction.

2.5 Microscope Objective

Equation (2.2) shows us that far-field diffraction is proportional to the Fourier transform of the incident wave if we adopt the modeling explained in the framework of the study.

We approximate that the microscope objective projects the diffraction pattern, which is supposed to be at infinity, onto its back focal plane. This approximation is valid because the objective transforms the diffracted plane waves from the object (which travel at a specific angle) into precise light points in its back focal plane.

This correspondence between the diffraction angle and the spatial position allows visualizing "far-field" diffraction at a finite distance.

Thus, we understand why the output field of the objective depends on the Fourier transform of the input.

Chapter 3

Numerical Approach

3.1 Implementation of the Fourier Transform

All simulations carried out in the project are based on the relation:

$$\vec{E}(x, y, z) = -\frac{if}{\lambda_0 k_t^2} \mathcal{F} \left[\frac{\vec{E}_t(\theta, \phi)}{\cos(\theta)} e^{ikz} \right] \quad (3.1)$$

At the center of this relation is the Fourier transform, calculated in our simulations using numpy's Fast Fourier Transform function. The input field of the pupil (E_t in the previous formula) is modeled and discretized by a grid of 250x250 pixels. The pupil is modeled by a disk of radius $R_{\text{pupille}} = 4\text{mm}$, giving a resolution of $32\mu\text{m}$ per pixel. Propagation along the optical axis is done on a grid 500 pixels wide, covering $[-100, 100]\mu\text{m}$, which corresponds to $0.4\mu\text{m}$ per pixel.

3.2 Calculation of the Propagated Field at the Optical System Output

The output field is calculated using the previous formula for a pre-selected list of z values. The result is a three-dimensional array of size $(N_x, N_y, N_z) = (250, 250, 500)$. Visualization of the calculated field is achieved by representing the 3 cross-sections (XY, XZ, YZ) passing through the spatial origin $(N_x/2, N_y/2, N_z/2)$.

3.3 Simulation Parameters

The physical parameters used during the simulations are as follows:

Parameters	Symbol	Value
Wavelength	λ	600nm
Objective focal length	f	5mm
Pupil radius	R_{pupille}	4mm
Numerical aperture	NA	0.8mm
Refractive index	n_t	1

Chapter 4

Simulation and Analysis

4.1 Gaussian Beam

4.1.1 Input Field in the Pupil

The incident field in the objective pupil is a Gaussian profile of the form:

$$E_t(r) = \exp\left(-\frac{r^2}{w_{\text{pupille}}^2}\right) \quad (4.1)$$

The field is centered on the optical axis and has central symmetry (around the spatial center of the image). The intensity decreases continuously from the center.

4.1.2 Field at the Focal Plane — XY Cross-Section

The field at the focal plane is obtained by the Fourier transform of the field in the pupil (Fraunhofer approximation), according to the relation:

$$\vec{E}(x, y, z) = -\frac{if}{\lambda_0 k_t^2} \mathcal{F}\left[\frac{\vec{E}_t(\theta, \phi)}{\cos(\theta)} e^{ikz}\right] \quad (4.2)$$

Since the Fourier transform of a Gaussian profile is a Gaussian, the field at the focal plane presents an intensity profile of the form:

$$I(r) = I_0 \exp\left(-\frac{r^2}{w_0^2}\right) \quad (4.3)$$

The XY cross-section confirms this behavior, we observe a compact lobe at the center, with circular symmetry (around the spatial center of the image). The theoretical focal plane radius is ($w_0 \approx 0.24$), which is consistent with the highly concentrated spot visible in the simulation.

4.1.3 Propagation — XZ and YZ Cross-Sections

The propagation cross-sections (XZ and YZ) show the characteristic structure of a Gaussian beam propagating, the beam radius is minimal at the focal plane ($z = 0$) then diverges on either side of the optical axis. The theoretical Rayleigh zone length is $z_R \approx 0.30 \mu\text{m}$ and defines the useful depth of field area of the system (the zone in which the beam has not yet diverged too much); beyond this, the beam radius evolves according to the relation:

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R}\right)^2} \quad (4.4)$$

It can be seen on the YZ and XZ cross-sections that the symmetry between the two transverse axes is well preserved, which agrees with the circular symmetry of the input profile.

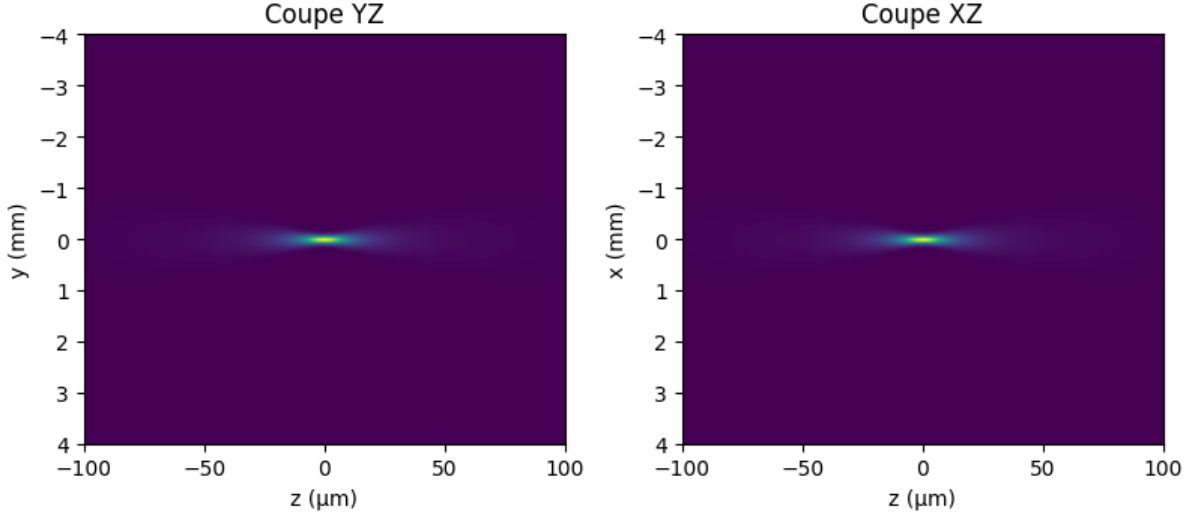


Figure 4.1: XZ and YZ transverse cross-section of the beam at the objective output

4.1.4 Summary

The Gaussian beam constitutes a reference case, it presents a minimal waist, but its short Rayleigh length ($z_R \approx 0.30$) reflects the trade-off between transverse confinement and axial extension. It is this limit that motivates the study of Bessel and Airy beams in the following sections.

4.2 Influence of the Numerical Aperture (NA) on the Waist Size at the Focal Plane

4.2.1 Theoretical Relation

The Full Width at Half Maximum (FWHM) of the Gaussian intensity profile at the focal plane is related to the numerical aperture by the relation:

$$\text{FWHM} = 2\sqrt{\ln 2} w_0 = 2\sqrt{\ln 2} \frac{\lambda f}{\pi R_{\text{pupille}}} = 2\sqrt{\ln 2} \frac{\lambda}{\pi \text{NA}} \quad (4.5)$$

This relationship predicts a $1/\text{NA}$ decrease, increasing the numerical aperture allows collecting higher frequency spatial components, which enables forming a thinner spot at the focal plane.

4.2.2 Simulation Results

Figure 4.2 compares the FWHM measured via numerical simulations (blue curve) to the theoretical prediction (orange curve) over a range from 0.016 to 0.8. Both curves follow the $1/\text{NA}$ law over the entire range of NA values, which validates the overall consistency of the simulation.

However, a discrepancy is observed at low NA values (typically $\text{NA} < 0.1$), where the practical curve overestimates the theoretical FWHM. Several numerical effects can explain this divergence:

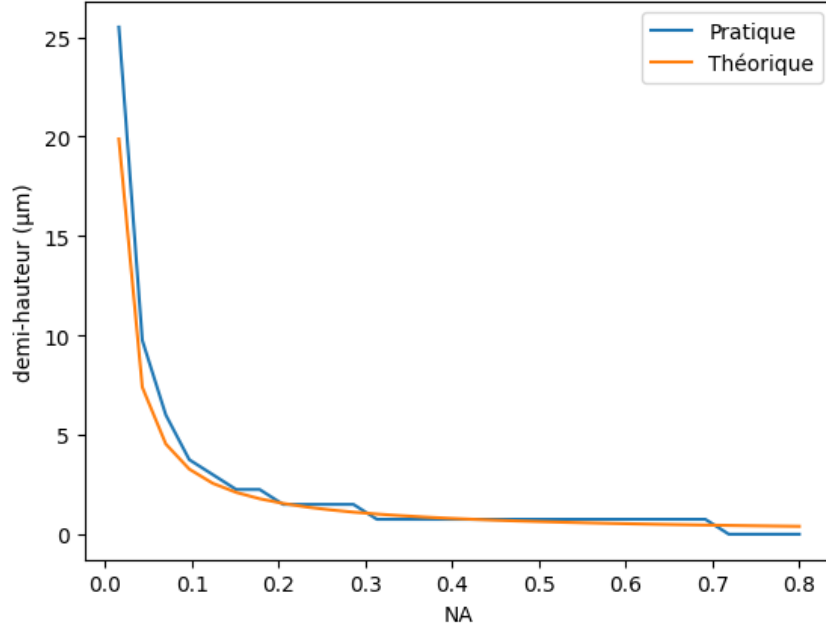


Figure 4.2: Comparison between theory and practice of the full width at half maximum of the Gaussian beam

- Insufficient grid resolution: for low NA values, the pupil occupies a reduced fraction of the simulation grid. The number of pixels describing the Gaussian profile decreases, which degrades the precision of the discrete Fourier transform and widens the calculated spot.
- Truncation and periodicity effects of the FFT: the discrete Fourier transform assumes signal periodicity. When the beam is highly spread out (low NA), its edges are no longer negligible at the grid boundaries, introducing aliasing effects that broaden the measured profile.

Conversely, for $NA > 0.3$, the two curves appear to overlap, which confirms that the simulation is reliable for larger NA values where the pupil is sufficiently well-sampled on the grid of points.

4.2.3 Summary

This result shows us the trade-off of light-sheet microscopy: a large numerical aperture improves transverse resolution (finer spot), but results in a shorter Rayleigh length, thus a reduced axial field of view. The simulation quantitatively confirms the law $FWHM \propto 1/NA$ with, however, numerical limitations for low values of the numerical aperture.

4.3 Gaussian Beam with Quadratic Phase : Simulation and Analysis

4.3.1 Input Field in the Pupil

The input field in the pupil has the same modulus as the previous beam; however, a quadratic phase modulated by an amplitude coefficient is added to it according to the form:

$$\phi = e^{-i\alpha(x^2+y^2)} \quad (4.6)$$

The amplitude α must be chosen so that the term in the exponential of the quadratic phase is at least of the same order of magnitude as the phase initially present in formula (1.1). This leads us to a coefficient $\alpha \approx 1e7$.

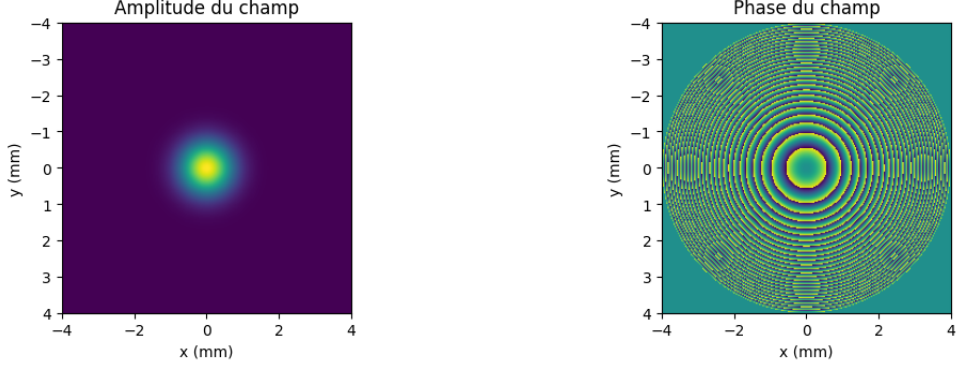


Figure 4.3: Amplitude and phase of the beam in the pupil plane of the objective

4.3.2 Field at the Focal Plane — XY Cross-Section

The field obtained in the focal plane is similar to that observed for a Gaussian beam without a quadratic phase. An output Gaussian intensity profile is obtained (thus maintaining central symmetry with respect to the spatial center of the image).

The radius at the focal plane remains identical to that of a simple Gaussian beam, so the beam has a focal plane radius of $w_0 \approx 0.24 \mu\text{m}$.

4.3.3 Propagation — XZ and YZ Cross-Sections

The main observable difference from the basic Gaussian beam is the defocusing appearing on the XZ and YZ cross-sections. Moreover, this defocusing seems at first glance to be proportional to the amplitude coefficient α .

The Rayleigh length remains identical to that of the Gaussian beam, as does the evolution of the beam radius if the new focal plane is treated as the origin of the z axis.

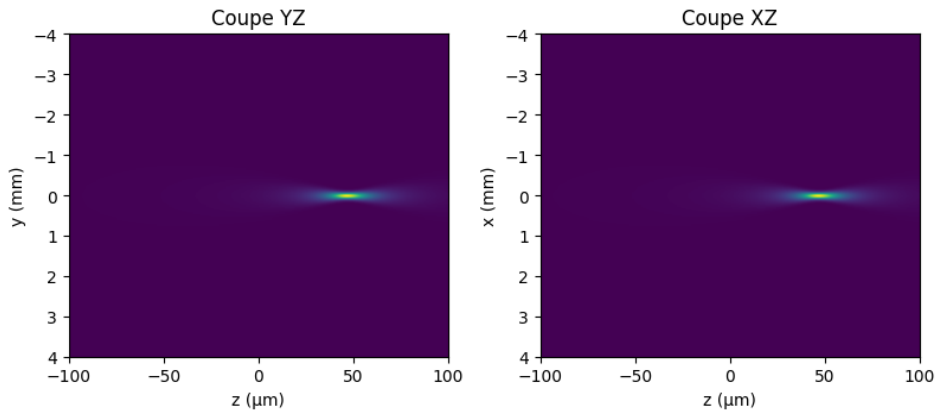


Figure 4.4: Shift of the object focal plane due to the quadratic phase added to the Gaussian beam

4.3.4 Empirical Modeling of Defocusing as a Function of α

The defocusing of the beam was measured for different values of the quadratic phase parameter α , yielding this graph.

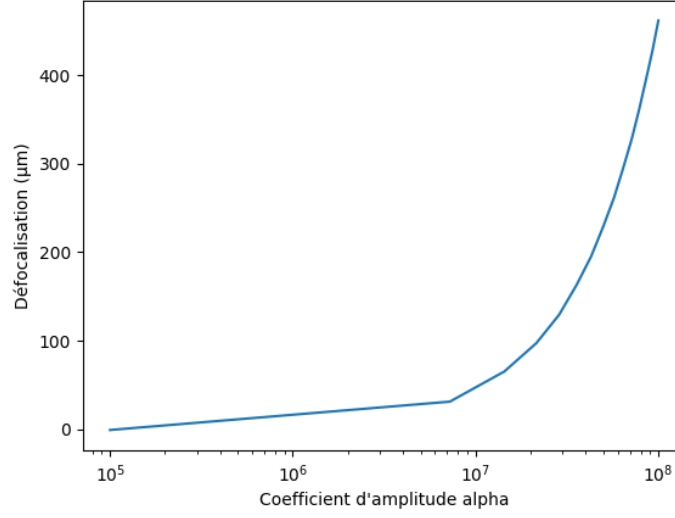


Figure 4.5: Shift of the focal plane as a function of the quadratic phase parameter α

Given the logarithmic scale, the defocusing of the beam appears to be linear to a power of α . A model of the following form was trained using the gradient descent method to find the optimal parameters:

$$y(\alpha) = a \cdot \alpha^b + c \quad (4.7)$$

The following results were found:

$$y(\alpha) = (3.264 \times 10^{-6}) \cdot \alpha^{1.019} \quad (4.8)$$

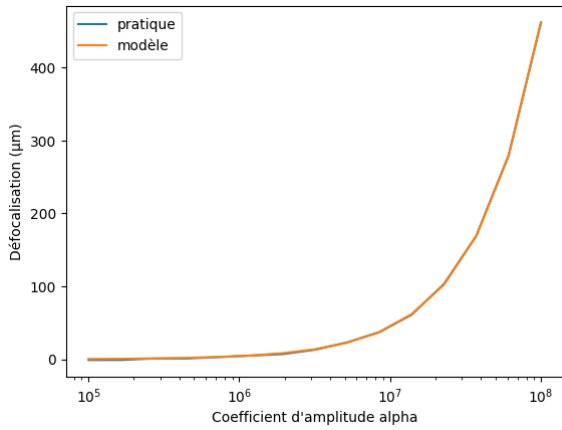


Figure 4.6: Comparison of the trained model with the practical defocusing curve as a function of α

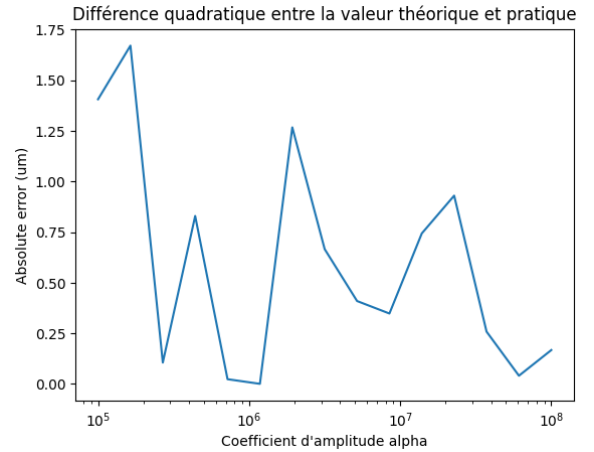


Figure 4.7: Absolute difference between the model and practice as a function of the value of α

The exponent $b = 1.019 \approx 1$ indicates that the defocusing is practically proportional to the amplitude coefficient α of the phase over the range $[1e5, 1e8]$. Furthermore, the model is quite close to the results obtained by simulation, keeping an error below $1.75\mu\text{m}$, even for the highest

values of α . It is noteworthy that the model performs better for high values of α . Indeed, although the defocusing increases, the error decreases and remains below a few μm .

4.3.5 Summary

The results show that adding a quadratic phase preserves the properties of the Gaussian beam, such as the size of its Rayleigh zone and the trade-off between resolution and axial extension. However, adding the quadratic phase allows controlling at what distance the image focal plane is located relative to the objective.

4.4 Bessel Beam — Simulation and Analysis

4.4.1 Input Field in the Pupil

An ideal Bessel beam is an exact solution of the Helmholtz equation of the form:

$$E(r, z) = J_0(k_r r) e^{ik_z z} \quad (4.9)$$

where J_0 is the zero-order Bessel function, and k_r , k_z are the radial and axial components of the wave vector. Such a beam is said to be non-diffracting, meaning its transverse profile is invariant along the axis of propagation (the z axis here).

In practice, an approximate Bessel beam is generated by restricting the field in the pupil to a thin ring of radius R_{anneau} and width δR :

$$E_t(r) = \begin{cases} 1 & \text{if } |r - R_{\text{anneau}}| \leq \delta R/2 \\ 0 & \text{otherwise} \end{cases} \quad (4.10)$$

4.4.2 Field at the Focal Plane — XY Cross-Section

The XY cross-section at the focal plane shows the structure of the J_0 function: a main lobe at the center surrounded by rings of decreasing intensity. Unlike the Gaussian beam, the energy is not solely concentrated at the center, but is also distributed across all secondary rings. The radius of the central lobe is given by the first zero of J_0 :

$$r_0 = \frac{2.405}{k_r} = \frac{2.405 f}{k R_{\text{anneau}}} \quad (4.11)$$

This profile comes from the fact that the Fourier transform of a ring is a Bessel function J_0 .

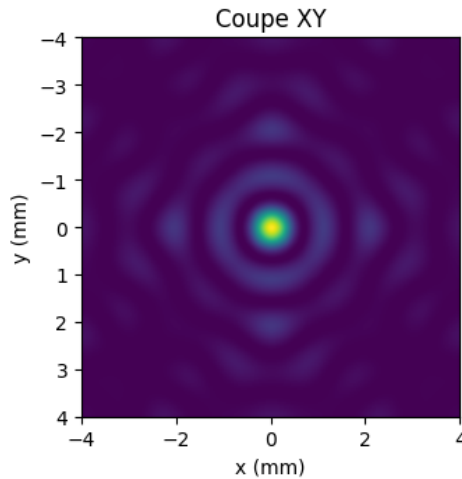
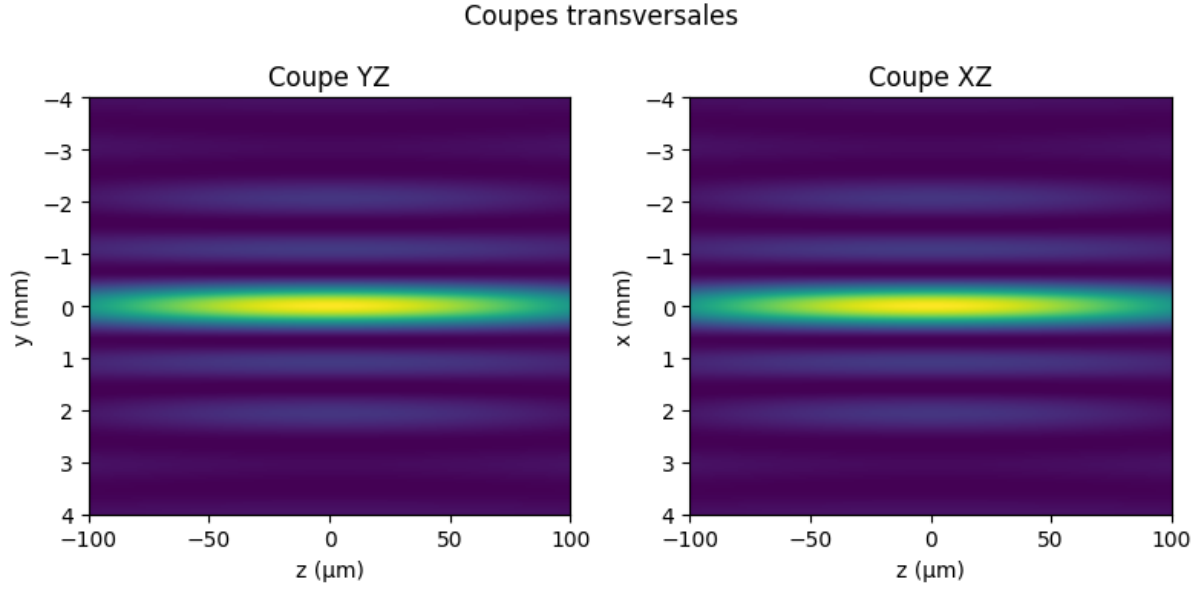


Figure 4.8: Field amplitude at the image focal plane of a Bessel beam

4.4.3 Propagation — XZ and YZ Cross-Sections



The propagation cross-sections (XZ and YZ) reveal the main characteristic of a propagating Bessel beam: the beam diffracts much slower than a basic Gaussian beam. Indeed, the ideal Bessel beam (its inner radius tends toward 0 and the ring width tends toward 0) has the characteristic of not diffracting, meaning its full width at half maximum remains constant along the optical axis (a result observed in Figure 4.9).

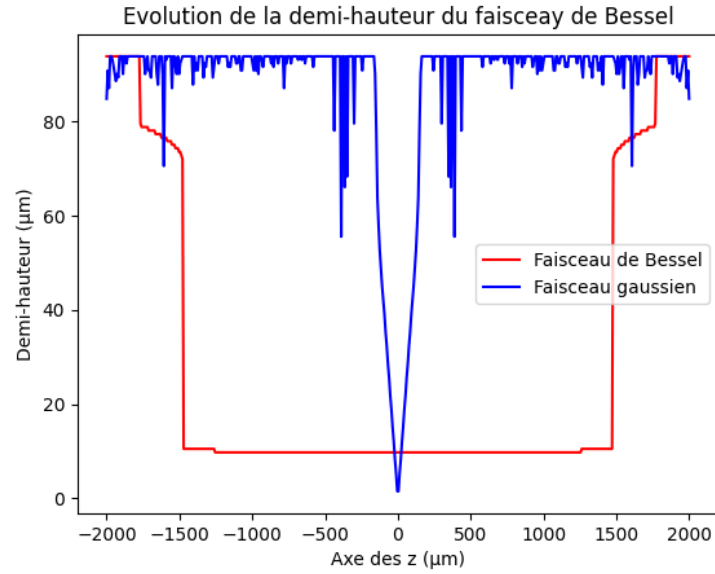


Figure 4.9: Comparison of the FWHM evolution of a Bessel beam and a Gaussian beam

The simulation whose results are presented in Figure 4.9 was carried out to come as close as possible to the ideal Bessel beam. The thickness of the ring as well as its radius were therefore reduced as much as possible.

We thus manage to obtain a zone of 3mm in size in which the Bessel beam will maintain a constant FWHM equal to about $10\text{ }\mu\text{m}$. On the other hand, the Gaussian beam has a FWHM that begins to increase right from the end of the Rayleigh zone (equal to $0.24\mu\text{m}$). We therefore

observe a zone in which the FWHM remains constant, about 12,000 times larger for the Bessel beam than for the Gaussian beam. Furthermore, by increasing the image resolution in the pupil plane, we can continue to decrease the radius and width of the ring, and thus get closer to the ideal Bessel beam (and thereby increase the size of the zone in which the FWHM remains constant).

4.4.4 Trade-off

The advantage gained in depth of field is lost against another counterpart. Indeed, a non-negligible fraction of the energy is distributed in the secondary rings rather than in the central lobe, which reduces image contrast in microscopy.

4.4.5 Summary

The Bessel beam offers a different trade-off than the Gaussian beam: in exchange for energy distributed over secondary rings, it maintains a central lobe of constant width over a much more extended depth of field than the Gaussian beam.

4.5 Airy Beam

4.5.1 Input Field in the Pupil

The incident field in the pupil has the same modulus as the previous beam, however, a cubic phase modulated by an amplitude coefficient is added to it according to the form:

$$\phi = e^{-i\alpha(x^3+y^3)} \quad (4.12)$$

The amplitude α must be chosen so that the term in the exponential of the quadratic phase is at least of the same order of magnitude as the phase initially present in formula (1.1). This leads us to a coefficient $\alpha \approx 5e9$.

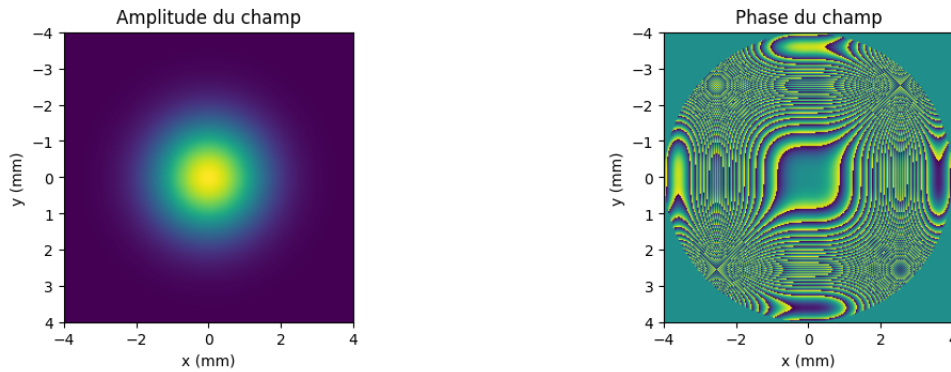


Figure 4.10: Amplitude of the Airy beam in the image focal plane

4.5.2 Field at the Focal Plane — XY Cross-Section

The field at the image focal plane presents a main lobe (lobe of maximum intensity) at the spatial center of the image having a square shape. Secondary lobes of rectangular shape are present around the main lobe following a rectangular tiling whose size varies according to the distance from the spatial center of the image.

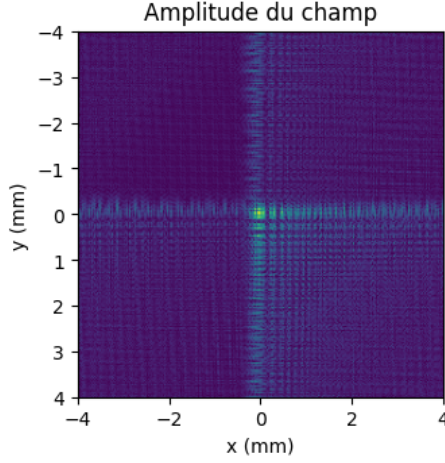


Figure 4.11: Beam amplitude in the pupil plane of the objective

4.5.3 Propagation — XZ and YZ Cross-Sections

The transverse cross-sections (XZ and YZ cross-sections) of the field at the objective output show one of the characteristics of the Airy beam: it propagates along a parabolic trajectory. Indeed, it propagates in these planes along a parabolic curve, as illustrated in the graphs below.

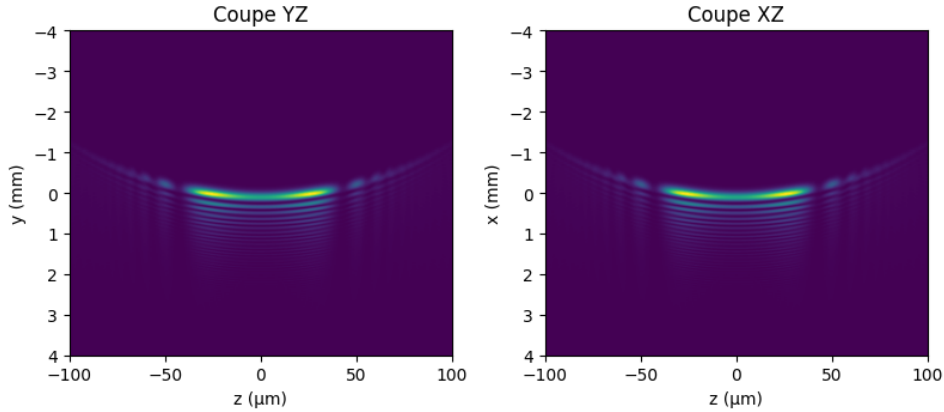


Figure 4.12: XZ and YZ cross-sections of the Airy beam at the objective output

Although small, a slight curvature is observed during the propagation of the beam. If we plot the position of the pixel of maximum intensity along the z axis, we obtain the graph of Figure 4.13.

We observe a variation of the trajectory along x of about $8\mu\text{m}$ (Figure 4.13) over a propagation length of $200\mu\text{m}$. Although slight, a parabolic propagation of the maximum intensity pixel is observed. A similar result is found if we look at the position along the y axis of the maximum intensity pixel (this is due to the central symmetry observed at the output of the objective).

4.5.4 Symmetries of the Airy Beam

The Airy beam, unlike other beams, does not exhibit rotational symmetry around the optical axis (z axis). Instead, the Airy beam exhibits symmetry with respect to the x and y axes separately. This is what gives its central lobe a square rather than a circular shape.

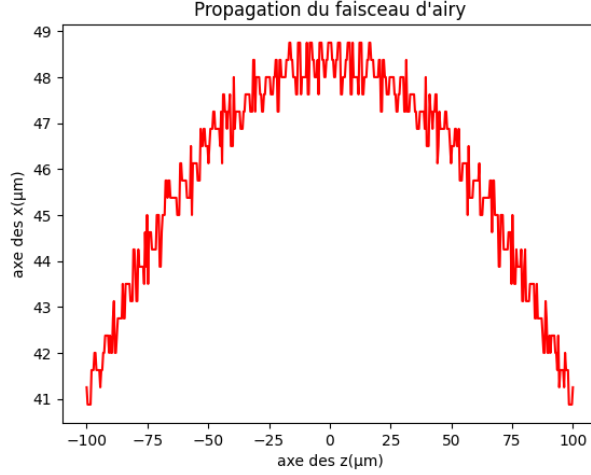


Figure 4.13: Evolution of the Airy beam trajectory by following the location of the maximum intensity pixel in the XZ cross-section (and thus YZ by symmetry)

4.6 Beam Regeneration Property

4.6.1 Simulation

One of the properties of the Airy beam is its self-healing capability. Indeed, if the beam encounters an obstacle during its propagation, it is capable of recovering a similar shape after a certain distance (which depends on the size of the obstacle). The energy lost when the beam hit the obstacle remains lost; however, the distribution of the proportion of energy in the XY plane returns to values similar to the initial ones.

To recover this result, we use a propagator called the Angular Spectrum Propagator (ASP) based on the following formula:

$$E(x, y, z) = \mathcal{F}^{-1} \{ \mathcal{F} \{ E(x, y, 0) \} \cdot H(f_x, f_y, z) \} \quad (4.13)$$

with:

$$H(f_x, f_y, z) = \begin{cases} \exp \left(ikz \sqrt{1 - (\lambda f_x)^2 - (\lambda f_y)^2} \right) & \text{if } f_x^2 + f_y^2 < \frac{1}{\lambda^2} \\ 0 & \text{otherwise} \end{cases} \quad (4.14)$$

To verify this property through simulations, we begin by creating an Airy beam using formula (1.1), which calculates the field at the objective output. We then place an obstacle at the spatial center of the image, right on the main lobe, and we calculate the propagated field for different values of z . This yields the results shown in Figure 4.14:

The field modeling the Airy beam was chosen as the field exiting the objective at the abscissa $z = 0\text{m}$, meaning at the image focal plane. As for the obstacle, it was modeled by a square whose dimensions were adjusted to completely cover the main lobe.

4.6.2 Theoretical Interpretation: Poynting Vector and Energy Flux

The physical explanation for the self-healing phenomenon of the Airy beam comes from the analysis of the Poynting vector, which describes the direction and density of the electromagnetic energy flux. Indeed, the Poynting vectors of the secondary lobes are directed toward the main lobe. Thus, when the energy of the main lobe is blocked by the obstacle, the energy from the secondary lobes moves toward the main lobe, depleting themselves in terms of energy until a

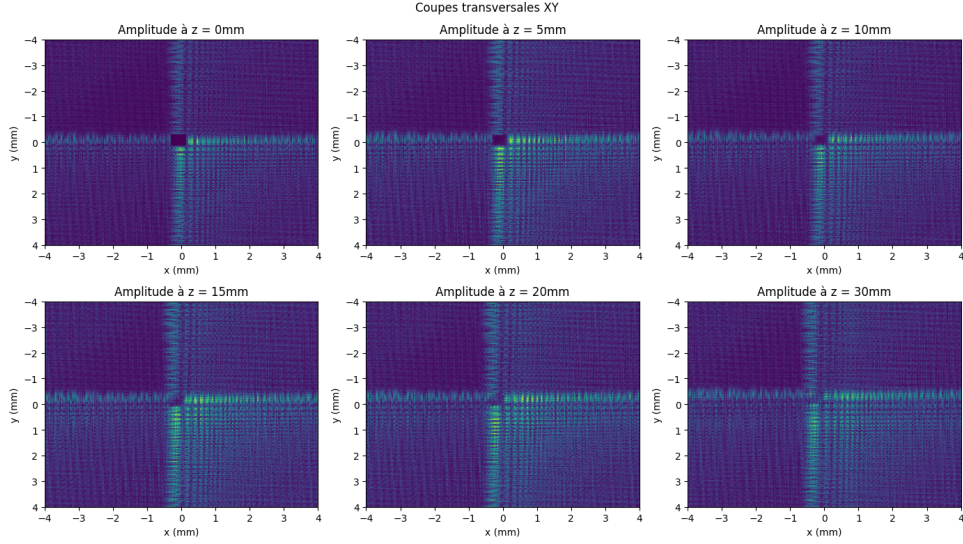


Figure 4.14: Evolution of the Airy beam intensity at several different values of z (the z -axis is along the optical axis) after encountering an obstacle

similar energy distribution is recovered. The energy of the main lobe is, however, permanently lost; only the energy distribution is preserved.

4.6.3 Summary

As with the Bessel beam, the energy of the Airy beam is distributed over many secondary lobes rather than being concentrated in the main lobe, which reduces imaging contrast. However, in light-sheet microscopy, the Airy beam is of particular interest for imaging samples containing obstacles or inhomogeneities, where its self-healing capability allows maintaining coherent illumination beyond local perturbations (biological imaging of thick tissues).

Chapter 5

Conclusion

The objective of this project was to simulate and numerically analyze Gaussian, Bessel, and Airy beams used in light-sheet microscopy by studying their properties.

The Gaussian beam (chosen as the reference standard) shows us the fundamental trade-off between a high resolution paired with a short depth of field (= size of the Rayleigh zone), or a lower resolution but a larger depth of field. The study of the FWHM verified that it is inversely proportional to the numerical aperture (NA).

The addition of a quadratic phase to the Gaussian beam highlighted the defocusing phenomenon, which is quasi-proportional to the coefficient α of the quadratic phase.

The Bessel beam, modeled by an annular profile, exhibits a nearly constant FWHM over several mm, which is approximately 12,500 times larger than the size of the Rayleigh zone (the area where the Gaussian beam is considered not to have diffracted too much yet). However, unlike the Gaussian beam, the energy of the Bessel beam is not concentrated solely in the central lobe but is distributed across its secondary rings in addition to the central lobe, thereby reducing imaging contrast.

The Airy beam exhibits properties that are vastly different from the first two. Indeed, in addition to propagating along a parabolic trajectory (a deviation of $8\mu\text{m}$ over $200\mu\text{m}$ of propagation), it possesses a self-healing capability that can be useful for imaging thick or inhomogeneous tissues. The explanation for this self-healing is due to the orientation of the Poynting vectors of the secondary lobes, which are directed toward the main lobe.

Furthermore, this project clearly demonstrates the limitations of numerical simulation. Indeed, throughout the simulations, numerical artifacts were observed (for instance, in the FWHM graph of the Gaussian beam, Figure 4.2). These artifacts can be attributed to the truncation effects of the Fourier transform. Other issues were encountered, for example, when measuring the FWHM of the Airy beam (likely due to the parabolic trajectory of the beam).

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